

SauceDemo — Naming Conventions & Sprint Breakdown

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PART 1: Naming Conventions & Traceability

Traceability means every artifact in your project — from a requirement to a test case to a bug report — can be **linked back to its source**. This is done through consistent naming/ID conventions.

1.1 The Traceability Chain

```
Business Need
├── User Story (US)
│   ├── Functional Requirement (FR)
│   │   ├── Test Case (TC)
│   │   │   ├── Cucumber Scenario (in .feature file)
│   │   │   │   └── Defect/Bug (BUG) ← if test fails
```

Every ID at each level **references the level above it**, so you can always trace a bug back to its original requirement.

1.2 Naming Convention Table

Artifact	Format	Example
User Story	US-[Sprint#]-[Story#]	US-01-03
Functional Requirement	FR-[Module#]-[Req#]	FR-01-04
Non-Functional Requirement	NFR-[Req#]	NFR-07
Test Case	TC-[ModuleCode]-[Seq#]	TC-LOGIN-003
Cucumber Scenario	[Action]_[Condition]_[ExpectedResult]	login_with_valid_credenti.
Step Definition Method	[action][Object][Condition]()	enterUsername()
Page Object Method	[verb][Element]()	clickLoginButton()
Defect/Bug	BUG-[ModuleCode]-[Seq#]	BUG-LOGIN-001
Feature File	[module_name].feature	login.feature

Artifact	Format	Example
Test Data Key	[module].[field]	login.validUsername

1.3 Module Codes

These short codes are used inside Test Case and Bug IDs:

Module	Code
Login / Authentication	LOGIN
Product Catalog	PROD
Product Detail	DETAIL
Shopping Cart	CART
Checkout	CHKOUT
Navigation Menu	NAV

1.4 Traceability Matrix Example

This table shows how everything links together:

User Story	FR ID	Test Case ID	Cucumber Scenario	Bug ID
US-01-01	FR-01-04	TC-LOGIN-001	login_valid_credentials_redirects_to_products	—
US-01-02	FR-01-05	TC-LOGIN-002	login_invalid_password_shows_error	—
US-01-03	FR-01-06	TC-LOGIN-003	login_empty_username_shows_required_error	—
US-01-04	FR-01-08	TC-LOGIN-004	login_locked_user_shows_locked_error	BUG-LOGIN-001
US-02-01	FR-02-01	TC-PROD-001	products_page_displays_six_products	—
US-03-01	FR-04-04	TC-CART-001	cart_remove_item_updates_badge_count	—

1.5 Cucumber Scenario Naming Rules

Scenarios in `.feature` files should follow this pattern:

```
[page/module]_[action]_[condition]_[expected result]
```

Examples:

```
gherkin
```

```
# ☒ Good – clear, traceable, reads like a sentence
```

```
Scenario: login_valid_credentials_redirects_to_products_page
```

```
Scenario: cart_remove_item_decrements_badge_count
```

```
Scenario: checkout_empty_firstname_shows_validation_error
```

```
# ☒ Bad – vague, not traceable
```

```
Scenario: Test login
```

```
Scenario: Check cart
```

```
Scenario: Form test
```

1.6 Java Naming Rules (for your framework)

```
java
```

```
// Page Object methods → verb + element
loginPage.enterUsername("standard_user");
loginPage.enterPassword("secret_sauce");
loginPage.clickLoginButton();
loginPage.getErrorMessage();

// Step Definition methods → describe the action
@When("the user enters valid credentials")
public void theUserEntersValidCredentials() { }

// Test data keys → module.field
config.get("login.validUsername")
config.get("login.password")
config.get("checkout.firstName")
```

PART 2: User Stories Broken into Sprints

Total Sprints: 5

Sprint Duration: 1 week each (recommended for a beginner)

Story Point Scale: 1 (trivial) → 5 (complex)

SPRINT 1 — Authentication & Login

Goal: Automate all login scenarios — the entry point of the application.

Feature File: `login.feature`

FR Coverage: FR-01-01 → FR-01-10

Story ID	User Story	FR Ref	Priority	Points
US-01-01	As a standard user, I want to log in with valid credentials so that I can access the products page	FR-01-04	High	2
US-01-02	As a user, I want to see an error when I enter an invalid password so that I know my credentials are wrong	FR-01-05	High	2
US-01-03	As a user, I want to see an error when I leave the username empty so that I know it is required	FR-01-06	High	1
US-01-04	As a user, I want to see an error when I leave the password empty so that I know it is required	FR-01-07	High	1
US-01-05	As a locked-out user, I want to see a clear locked-out message so that I understand why I cannot log in	FR-01-08	High	2
US-01-06	As a user, I want to dismiss the error message by clicking (X) so that the form is clean again	FR-01-09	Medium	1

Sprint 1 Total Points: 9

Acceptance Criteria Example (US-01-01):

gherkin

Feature: **Login**

Scenario: **login_valid_credentials_redirects_to_products_page**

Given the user is on the login page

When the user enters username "standard_user"

And the user enters password "secret_sauce"

And the user clicks the login button

Then the user should be redirected to the products page

And the page title should display "Products"

SPRINT 2 — Product Catalog

Goal: Automate product listing, sorting, and add-to-cart behavior.

Feature File: `product_catalog.feature`

FR Coverage: FR-02-01 → FR-02-10

Story ID	User Story	FR Ref	Priority	Points
US-02-01	As a shopper, I want to see all 6 products on the catalog page so that I can browse available items	FR-02-01	High	2
US-02-02	As a shopper, I want each product to show a name, image, description, and price so that I can make informed choices	FR-02-02	High	2
US-02-03	As a shopper, I want to add a product to the cart so that I can purchase it later	FR-02-03, FR-02-04	High	2
US-02-04	As a shopper, I want the "Add to cart" button to change to "Remove" after clicking so that I know the item is in my cart	FR-02-04	High	1
US-02-05	As a shopper, I want the cart badge to update when I add items so that I always know how many items are in my cart	FR-02-06	High	2
US-02-06	As a shopper, I want to sort products by name (A-Z and Z-A) so that I can find products alphabetically	FR-02-07	Medium	2
US-02-07	As a shopper, I want to sort products by price (low-high and high-low) so that I can find items in my budget	FR-02-07	Medium	2
US-02-08	As a shopper, I want to click a product name to view its detail page so that I can read more information	FR-02-08	Medium	1

Sprint 2 Total Points: 14

Acceptance Criteria Example (US-02-03):

gherkin

Scenario: **catalog_add_product_to_cart_updates_badge**

Given the user is logged in as "standard_user"

And the user is on the products page

When the user clicks "Add to cart" on "Sauce Labs Backpack"

Then the button text should change to "Remove"

And the cart badge should display "1"

SPRINT 3 — Shopping Cart

Goal: Automate cart management — adding, removing, and navigating from the cart.

Feature File: `shopping_cart.feature`

FR Coverage: FR-04-01 → FR-04-09

Story ID	User Story	FR Ref	Priority	Points
US-03-01	As a shopper, I want to view all items I added in the cart so that I can review my selection before buying	FR-04-01, FR-04-02	High	2
US-03-02	As a shopper, I want to remove an item from the cart so that I can change my mind before checkout	FR-04-03, FR-04-04	High	2
US-03-03	As a shopper, I want the cart badge to decrease when I remove an item so that the count stays accurate	FR-04-04	High	1
US-03-04	As a shopper, I want to click "Continue Shopping" so that I can go back and add more products	FR-04-05, FR-04-06	High	1
US-03-05	As a shopper, I want to click "Checkout" so that I can proceed to complete my purchase	FR-04-07, FR-04-08	High	2
US-03-06	As a shopper, I want an empty cart to show no items and no errors so that I am not confused	FR-04-09	Medium	1

Sprint 3 Total Points: 9

Acceptance Criteria Example (US-03-02):

gherkin

Scenario: `cart_remove_item_decrements_badge_count`

Given the user is logged in as "standard_user"

And the user has added "Sauce Labs Backpack" to the cart

And the user navigates to the cart page

When the user clicks "Remove" on "Sauce Labs Backpack"

Then the item should no longer appear in the cart

And the cart badge should not be displayed

SPRINT 4 — Checkout Flow

Goal: Automate the complete checkout process — information form, order overview, and confirmation.

Feature File: `checkout.feature`

FR Coverage: FR-05-01 → FR-07-05

Story ID	User Story	FR Ref	Priority	Points
US-04-01	As a shopper, I want to complete checkout with valid information so that my order is placed successfully (happy path)	FR-05-02, FR-06-06, FR-07-01	High	3
US-04-02	As a shopper, I want to see an error when I leave First Name empty so that I know it is required	FR-05-03	High	1
US-04-03	As a shopper, I want to see an error when I leave Last Name empty so that I know it is required	FR-05-04	High	1
US-04-04	As a shopper, I want to see an error when I leave Zip Code empty so that I know it is required	FR-05-05	High	1
US-04-05	As a shopper, I want to see the item total, tax, and final price on the order overview so that I can confirm costs before paying	FR-06-02, FR-06-03, FR-06-04	High	2
US-04-06	As a shopper, I want to see a success confirmation after finishing checkout so that I know my order was placed	FR-07-01, FR-07-02	High	2
US-04-07	As a shopper, I want the cart badge to be cleared after a completed order so that the app resets correctly	FR-07-05	High	2
US-04-08	As a shopper, I want to click "Back Home" after checkout so that I return to the products page	FR-07-04	Medium	1
US-04-09	As a shopper, I want to cancel checkout step 1 and return to the cart so that I can reconsider my items	FR-05-06	Medium	1

Sprint 4 Total Points: 14

Acceptance Criteria Example (US-04-01):

gherkin

Scenario: **checkout_happy_path_completes_successfully**

Given the user is logged in as "standard_user"

And the user has added "Sauce Labs Backpack" to the cart

And the user is on the cart page

When the user clicks "Checkout"

And the user enters first name "John", last name "Doe", zip code "12345"

And the user clicks "Continue"

And the user clicks "Finish"

Then the confirmation header should display "Thank you for your order!"

And the cart badge should not be visible

SPRINT 5 — Navigation, Edge Cases & Problem Users

Goal: Automate the hamburger menu, logout, reset state, and behavior differences across user types.

Feature File: `navigation.feature`, `user_behavior.feature`

FR Coverage: FR-08-01 → FR-08-06, NFR-06, NFR-07

Story ID	User Story	FR Ref	Priority	Points
US-05-01	As a logged-in user, I want to open and close the hamburger menu so that I can access navigation options	FR-08-01, FR-08-06	High	1
US-05-02	As a logged-in user, I want to click "Logout" so that I am returned to the login page and my session ends	FR-08-03	High	2
US-05-03	As a logged-in user, I want to click "All Items" so that I am navigated to the products page	FR-08-02	High	1
US-05-04	As a logged-in user, I want to click "Reset App State" so that the cart is cleared and buttons reset	FR-08-04	Medium	2
US-05-05	As a QA engineer, I want to verify that unauthenticated users cannot access <code>/inventory.html</code> directly so that the app is secure	NFR-07	High	2
US-05-06	As a QA engineer, I want to verify that the <code>problem_user</code> experiences broken images so that known defects are documented	FR-02-02	Medium	3
US-05-07	As a QA engineer, I want to verify that the <code>performance_glitch_user</code> has a slower login so that performance issues are captured	NFR-01	Medium	3

Sprint 5 Total Points: 14

Acceptance Criteria Example (US-05-02):

gherkin

Scenario: `navigation_logout_redirects_to_login_page`

Given the user is logged in as "standard_user"

When the user opens the navigation menu

And the user clicks "Logout"

Then the user should be on the login page

And the URL should be "https://www.saucedemo.com/"

Sprint Summary

Sprint	Focus Area	Stories	Total Points
Sprint 1	Authentication & Login	6	9
Sprint 2	Product Catalog	8	14
Sprint 3	Shopping Cart	6	9
Sprint 4	Checkout Flow	9	14
Sprint 5	Navigation & Edge Cases	7	14
Total		36 stories	60 points

Recommended Sprint Order (Why This Order?)

Sprint 1 (Login) → Must come first; all other sprints depend on being logged

in

Sprint 2 (Catalog) → Must come before cart; you add items from here

Sprint 3 (Cart) → Must come before checkout; checkout starts from cart

Sprint 4 (Checkout) → Full end-to-end flow; requires Sprints 1-3

Sprint 5 (Navigation) → Can run independently; adds depth and edge case coverage